



Simulations of counterflow dual-fuel flames and implications for DCMC

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ABSTRACT

In anticipation of future multi-dimensional turbulent combustion calculations of dual-fuel systems with the Doubly Conditional Moment Closure (DCMC), a preliminary validation of the DCMC-calculated structures of such flames in mixture fraction – progress variable space has been performed by comparison with simulations of laminar non-premixed counterflow flames of gaseous heptane against methane/air mixtures as the oxidizer under various conditions. Two methane/air equivalence ratios were considered, one below and one slightly above the lean flammability limit of the methane/air premixed mixture, the former selected as a particularly challenging case where the interaction between the two fuels is expected to be strong since the methane-air flame cannot sustain on its own. The effect of strain was also investigated. The accuracy of the usual assumption of unity Lewis number employed in the DCMC model was examined using the laminar flame simulations and it was concluded that some discrepancies exist in intermediate species and temperature profiles compared to the case when differential diffusion was taken into account. These differences were enhanced at high strain. The heat release rate profile implies the existence of two separate reaction zones, which merge at high strain rates. The DCMC model predicted quite accurately the methane/air side of the flame in mixture fraction space, but not to the same degree of accuracy the heptane side of the flame. The results provide insights into the structure of dual-fuel flames and suggest that such systems can be successfully modelled by mixture-fraction based methods.

1. Introduction

Dual-fuel combustion systems are increasingly considered for a wide range of applications [1–3]. A model that is capable of capturing the turbulence-chemistry interactions at all scales and for all modes of combustion is required to model the behavior of these systems.

One of the potential methods is the Conditional Moment Closure (CMC) approach for turbulent reacting flows [3]. This method derives transport equations for the conditional averages of enthalpy and species mass fractions based on one or more sample space parameters (mixture fraction for non-premixed flames or reaction progress variable for premixed flames or both as in this paper). CMC can use detailed chemistry and includes micro- and macro-mixing [4]. It can also be coupled with Large Eddy Simulations. CMC-LES has been validated and employed for a wide range of combustion problems such as ignition [5], gaseous flame structure [6], and flames (premixed or non-premixed) close to blow-off [7].

In all the aforementioned cases, a single parameter is enough to capture the thermochemistry under study, but this is not the case for

practically relevant dual-fuel flames. In these cases, an increased number of conditioning parameters is required. Considering all the above, Doubly Conditional Moment Closure (DCMC) models have been developed but mainly only been applied at the level of fundamental feasibility studies [8–10]. A successful practical simulation model (including an LES version of it) has been developed by Sitte and Mastorakos [11–13] for the simulation of spray flames with the choice of a mixture fraction and a reaction progress variable as the two conditioning parameters. Their study has shown that DCMC can capture multi-modal combustion. In the present work, DCMC is used for two fuels systems.

This paper has two main objectives: Firstly, to explore the canonical dual-fuel combustion problem of a diffusion counterflow flame, in which C_7H_{16} and a premixed CH_4 /air mixture are the two opposing streams using the DCMC method to examine the capabilities and some limitations of this model by comparing its results with predictions of Cantera [14]. This composition has been experimentally studied in a swirling C_7H_{16} - CH_4 /air flame by Sidey and Mastorakos [15]. Secondly, to analyze, using Cantera laminar flame simulations, the detailed structures of such non-premixed counterflow flames at various strain rates.

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2. Methods

2.1. DCMC and OD-DCMC

DCMC extends the CMC concept, which has a single conditioning parameter ξ , with corresponding sample space variable η , with the introduction of a second conditioning parameter c with corresponding sample space variable ζ [3]. DCMC derives transport equations for the conditional averages:

$$Q_\alpha(\eta, \zeta) = \langle \varphi(\mathbf{x}, t) | \xi(\mathbf{x}, t) = \eta, c(\mathbf{x}, t) = \zeta \rangle \quad (1)$$

Here Q_α is the conditional average of the reactive scalar α , $\mathbf{x}$ is the position vector, and t is the time. ξ is the mixture fraction which takes the value of unity ($\xi = 1$) at the locations where pure C_7H_{16} is found, and the value of zero ($\xi = 0$) at the CH_4 /air premixed mixture. c is the reaction progress variable defined as unity ($c = 1$) when locally chemical equilibrium has been achieved and zero ($c = 0$) if no reaction has taken place. It can be defined based on a linear combination of species mass fractions Y_ψ [11] which are functions of the local mixture fraction:

$$c(\xi) = [Y_\psi^0(\xi) - Y_\psi(\xi)] / [Y_\psi^0(\xi) - Y_\psi^{eq}(\xi)] \quad (2)$$

where the superscripts “0” and “eq” denote the mixing line and the equilibrium state respectively. Here Y_ψ was defined as the sum of the mass fractions of CO_2 , CO and H_2O so that the reaction progress variable increases monotonically and takes values only in the range between 0 and 1.

“OD-DCMC” is the solution of DCMC under the special conditions of spatial homogeneity and with prescribed scalar dissipation rate distributions in the $\eta - \zeta$ space. The solutions serve as reference and offer insights into the process. The OD-DCMC equations for species mass fraction α and enthalpy h can be found in [13] and are given below:

$$\begin{aligned} \frac{\partial Q_\alpha}{\partial t} = & \langle N_\xi | \eta, \zeta \rangle \frac{\partial^2 Q_\alpha}{\partial \eta^2} + 2 \langle N_{\xi c} | \eta, \zeta \rangle \frac{\partial^2 Q_\alpha}{\partial \eta \partial \zeta} \\ & + \langle N_c | \eta, \zeta \rangle \frac{\partial^2 Q_\alpha}{\partial \zeta^2} + \langle \dot{\omega}_\alpha | \eta, \zeta \rangle - \langle \omega_c^* | \eta, \zeta \rangle \frac{\partial Q_\alpha}{\partial \zeta} \end{aligned} \quad (3)$$

$$\begin{aligned} \frac{\partial Q_h}{\partial t} = & \langle N_\xi | \eta, \zeta \rangle \frac{\partial^2 Q_h}{\partial \eta^2} + 2 \langle N_{\xi c} | \eta, \zeta \rangle \frac{\partial^2 Q_h}{\partial \eta \partial \zeta} \\ & + \langle N_c | \eta, \zeta \rangle \frac{\partial^2 Q_h}{\partial \zeta^2} - \langle \omega_c^* | \eta, \zeta \rangle \frac{\partial Q_h}{\partial \zeta} \end{aligned} \quad (4)$$

where $\langle \omega_c^* | \eta, \zeta \rangle$ is the conditional apparent reaction rate given by:

$$\begin{aligned} \langle \omega_c^* | \eta, \zeta \rangle = & \frac{1}{\partial Q_\psi / \partial \zeta} (\langle \dot{\omega}_\psi | \eta, \zeta \rangle + \langle N_\xi | \eta, \zeta \rangle \frac{\partial^2 Q_\psi}{\partial \eta^2} \\ & + 2 \langle N_{\xi c} | \eta, \zeta \rangle \frac{\partial^2 Q_\psi}{\partial \eta \partial \zeta} + \langle N_c | \eta, \zeta \rangle \frac{\partial^2 Q_\psi}{\partial \zeta^2}) \end{aligned} \quad (5)$$

In Eqs. (3)–(5), $\dot{\omega}_\varphi$ is the production rate of the reactive scalar φ , N_ξ and N_c denote the scalar dissipation rates of mixture fraction and reaction progress variable respectively, and $N_{\xi c}$ is the cross-scalar dissipation rate. The scalar dissipation rates are defined by:

$$N_\xi = D \left(\frac{\partial \xi}{\partial x} \right)^2, \quad N_c = D \left(\frac{\partial c}{\partial x} \right)^2, \quad N_{\xi c} = D \frac{\partial \xi}{\partial x} \frac{\partial c}{\partial x} \quad (6)$$

where D is the diffusion coefficient. Note that the above formulation assumes that the Lewis number (Le) takes the value of unity. $\langle \dot{\omega}_\varphi | \eta, \zeta \rangle$ was modelled using a first order closure assuming that the fluctuations are small [11]. $\langle N_\xi | \eta, \zeta \rangle$ was modelled using the amplitude mapping closure (AMC) bell-shaped curve [13]

$$\langle N_\xi | \eta, \zeta \rangle = N_{\xi, \max} \exp \left(-2 [\text{erf}^{-1}(2\eta - 1)]^2 \right) \quad (7)$$

It was assumed that the cross-scalar dissipation rate was zero. For $\langle N_{\xi c} | \eta, \zeta \rangle$, results obtained from the simulation of premixed counterflow

Table 1

Maximum scalar dissipation rate values and global strain rate of the simulated flames.

Simulation [φ]+[strain]	$N_{\xi, \max} [s^{-1}]$	$a [s^{-1}]$	$N_{c, \max} [s^{-1}]$
PHI0.14LOW	32.94	103.48	1598.13
PHI0.14HIGH	98.82	310.44	1598.13
PHI0.56LOW	49.70	156.13	1480.76
PHI0.56HIGH	129.21	405.93	1480.76

flames (mixture against its equilibrium products) with a strain rate of $5 s^{-1}$ were employed. At each η , the mixture was formed by the combination of C_7H_{16} and CH_4 /air in proportions η and $1 - \eta$ by mass respectively. For the mixture fraction values for which a burning solution could not be found, a value of zero was assumed for the scalar dissipation rate of the reaction progress variable. The above modelling assumptions have been employed in an LES-DCMC simulation of the C_7H_{16} - CH_4 /air Cambridge swirl flame [15] with reasonable success [16]. Eqs. (3)–(5) were here solved using the CLIO code [17,18] that uses operational splitting with fractional steps and the DVODPK solver. The simulations were initialized by assuming gradual progress towards the full combustion of the reactants (i.e. full conversion to CO_2 and H_2O , without the formation of intermediate species) profile over the $\eta - \zeta$ space for all conditional quantities and giving priority to the combustion of CH_4 over that of C_7H_{16} . For the discretization of the $(\eta - \zeta)$ space, 71 points were chosen along the η dimension following a beta distribution and 51 uniformly distributed points were chosen along the c dimension. Pressure was assumed to be 1 atm, the inlet temperature of C_7H_{16} (at $\eta = 1$) was set equal to the boiling temperature of heptane ($T_{boil} = 371.5 K$) [19] and the inlet temperature of the CH_4 /air premixed mixture (at $\eta = 0$) with a CH_4 /air equivalence ratio of $\varphi = 0.14$ or $\varphi = 0.56$ was $T_{pmx} = 300 K$ to be consistent with the CH_4 /air premixed mixtures studied experimentally [15]. At $\zeta = 0$ and $\zeta = 1$ inert mixing line and equilibrium conditions were enforced respectively. The values of the maxima of the scalar dissipation rates used in the simulations are shown in Table 1.

2.2. Cantera simulations

The counterflow flames of Table 1 were solved using Cantera using both a Le=1 transport model and a mixture averaged transport model. Two equivalence ratios, $\varphi = 0.14, 0.56$ were considered as in the turbulent swirl flame experiments [15]. The detailed mechanism of Smallbone et al. [20] was employed. A comparison with experimental data for laminar burning velocities of freely propagating flames of CH_4 and C_7H_{16} separately, and for the extinction strain rates of C_7H_{16} non-premixed flames [21–23] was undertaken. Fig. S1 in Supplementary Material shows the predicted and experimentally measured values of laminar burning velocity (S_L) for both C_7H_{16} and CH_4 .

Concerning the extinction strain rate of non-premixed flames, nitrogen-diluted C_7H_{16} diffusion flames at various mass fractions $Y[C_7H_{16}, f]$ in the fuel stream (f) at a temperature of $T_f = 533 K$ against a pure air oxidizer stream of temperature $T_{ox} = 533 K$ were measured [21]. The extinction strain rate was determined by:

$$k_{ext, ox} = \frac{2V_{ox, ext}}{w} \left(1 + \frac{V_{f, ext} \sqrt{\rho_f}}{V_{ox, ext} \sqrt{\rho_{ox}}} \right) \quad (8)$$

where $V_{f, ext}$, ρ_f and $V_{ox, ext}$, ρ_{ox} are the normal velocities to the flame plane at extinction and the densities of the fuel and oxidizer streams respectively, and w is the distance between the nozzles. In the Cantera simulations, the momenta of the two streams were set equal and a nozzle distance, $w = 0.01 m$ was chosen. The global strain rate (a) was defined by:

$$a = (V_f + V_{ox})/w \quad (9)$$

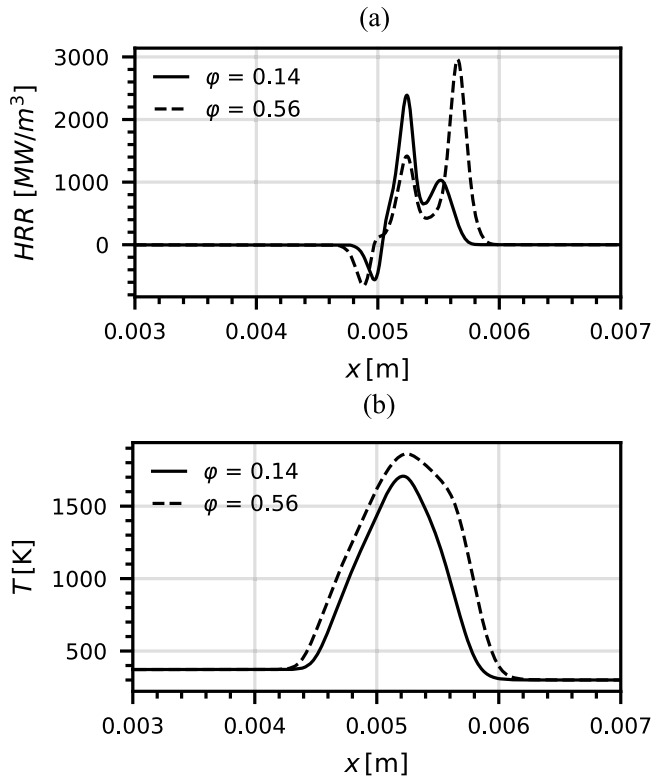


Fig. 1. Heat release rate (HRR) and temperature profiles of counterflow non-premixed flames with a strain rate of $a = 310 \text{ s}^{-1}$ and the indicated CH_4/air equivalence ratio.

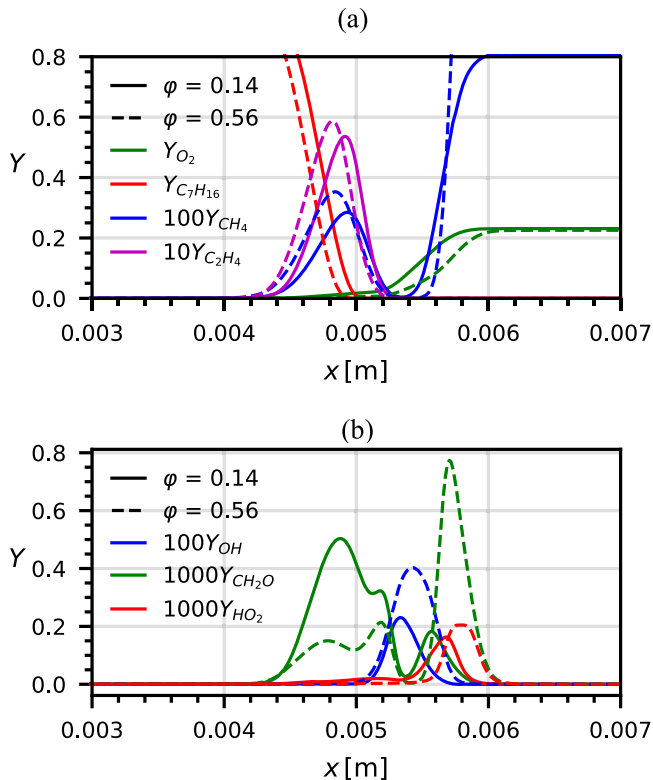


Fig. 2. Species mass fraction profiles of counterflow non-premixed flames with a strain rate of $a = 310 \text{ s}^{-1}$ and the indicated CH_4/air equivalence ratio.

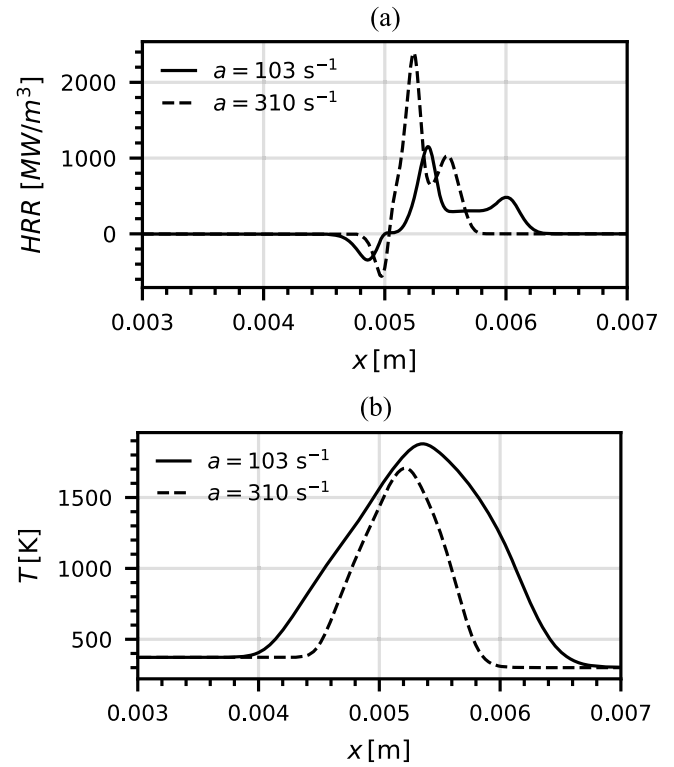


Fig. 3. Heat release rate and temperature profiles of counterflow non-premixed flames with CH_4/air equivalence ratio of $\phi = 0.14$ and the indicated strain rate.

Table 1 includes the strain rate values computed using Eq. (9).

The heptane non-premixed flame extinction strain rates from the experiment [21] and the simulations are shown in Fig. S2 of the Supplementary Material. The good agreement between the predictions and the experimental results suggest that this detailed mechanism should be sufficient for the dual-fuel problem under investigation.

3. Results and discussion

In this Section, the results of the Cantera simulations are first presented and discussed, followed by simulations with the DCMC solver to assess the accuracy of flame structures as captured by DCMC.

3.1. Laminar flame structures from Cantera

Non-premixed flames whose oxidizer stream was a CH_4/air premixed mixture of $\phi = 0.14$, $\phi = 0.47$, (below the CH_4/air lean flammability limit of $\phi_{\text{LFL}} = 0.5$ [24]), and $\phi = 0.56$ (above the CH_4/air lean flammability limit) extinguished approximately at $a \approx 330 \text{ s}^{-1}$, $a \approx 425 \text{ s}^{-1}$ and $a \approx 415 \text{ s}^{-1}$ respectively. Therefore, the addition of CH_4 in the oxidizer initially causes a slight increase in the extinction strain rate as the CH_4/air lean flammability limit is approached and then causes a decrease. The manifestation of a maximum can be attributed to the competition of the two fuels (CH_4 and C_7H_{16}) for the consumption of oxygen. The structure of flames of the two equivalence ratios under consideration with $a = 310 \text{ s}^{-1}$ can be seen in Figs 1-2.

The two HRR maxima in Fig. 1a correspond to the oxidation of CH_4 (right maximum) and C_7H_{16} (left maximum). At a higher ϕ , the maximum value of the CH_4 oxidation region increases while the maximum of the C_7H_{16} oxidation region decreases.

It is evident that as ϕ increases the temperature increases everywhere (Fig. 1b). The greatest increase occurs in the CH_4/air consistent with the change observed in the HRR profile. The C_7H_{16} , CH_4 , O_2 and C_2H_4 mass fraction distributions are presented in Fig. 2a. On the C_7H_{16} side, C_2H_4

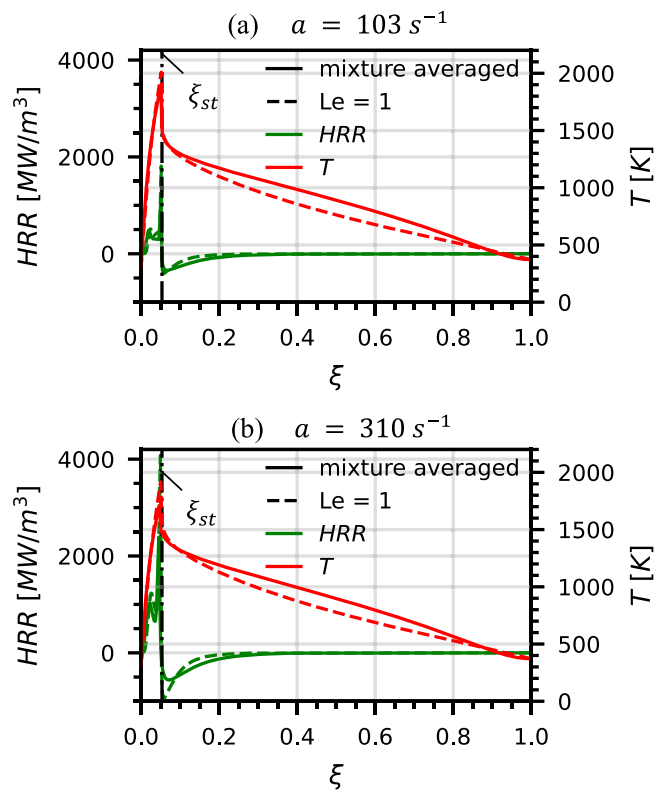


Fig. 4. Heat release rate and temperature profiles of counterflow non-premixed flames with $\varphi = 0.14$ and strain rate as indicated. The dashed vertical line shows the stoichiometric mixture fraction.

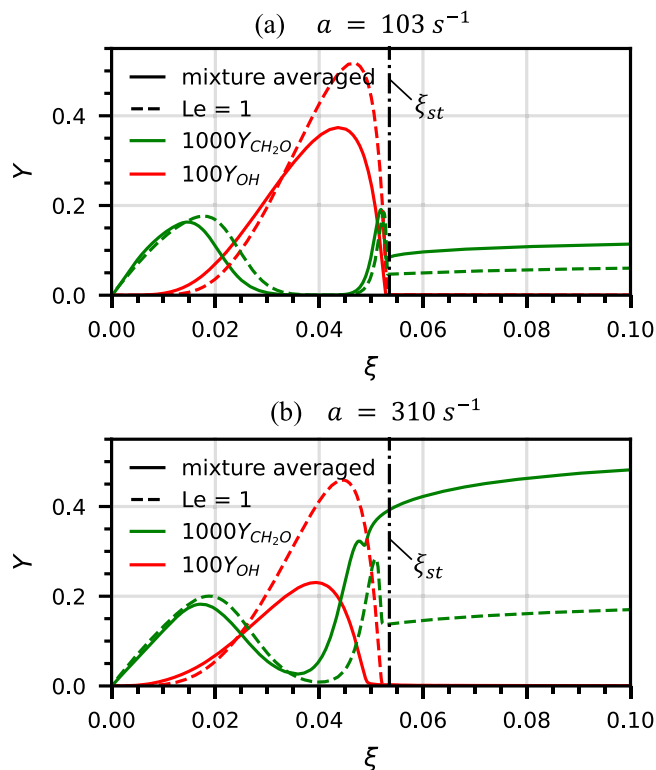


Fig. 5. CH_2O and OH mass fraction profiles of counterflow non-premixed flames with $\varphi = 0.14$ and strain rate as indicated. The dashed vertical line shows the stoichiometric mixture fraction.

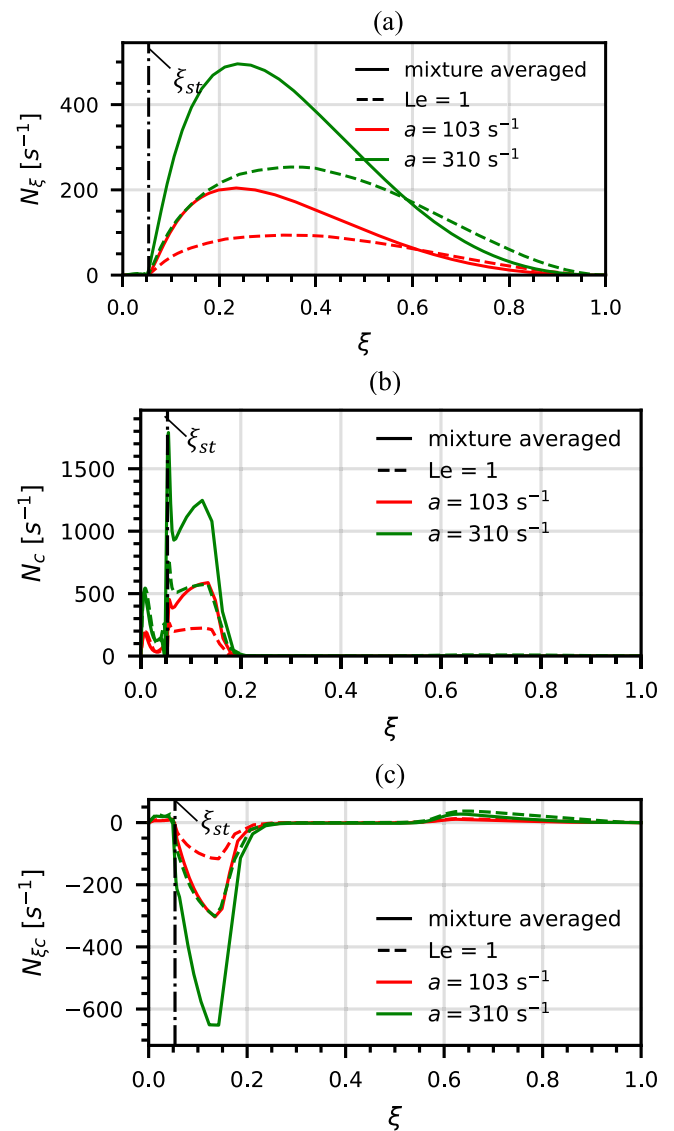


Fig. 6. Scalar dissipation profiles of counterflow non-premixed flames with $\varphi = 0.14$ and the strain rates indicated. The dashed vertical line shows the stoichiometric mixture fraction.

forms a peak at the location where CH_4 is present, which is also the location where the HRR becomes negative. This is expected because on the C_7H_{16} side endothermic pyrolysis of the heptane takes place producing smaller hydrocarbons.

Fig. 2b shows also the profiles of CH_2O , OH and HO_2 . At the low φ , CH_2O is mainly formed on the C_7H_{16} side, while at a greater φ it is mainly formed on the CH_4/air side because of the significantly enhanced combustion of CH_4 at this equivalence ratio (which is above the lean flammability limit of CH_4/air premixed mixture). OH is mainly formed where temperature takes its highest values. In $\varphi = 0.56$ case, the thickness of the OH bell-shaped curve increases significantly and so does the distance between the two CH_2O regions. HO_2 radicals participate in termination steps, but they are also involved in the production of radicals at high temperatures [25]. At the higher φ , the region of their main presence is displaced towards the CH_4/air side and their mass fraction there increases but their “leakage” decreases.

Fig. 3 and Fig. S3 in the Supplementary Material show results for counterflow non-premixed flames of the $\text{C}_7\text{H}_{16}\text{-CH}_4/\text{air}$ dual-fuel system with $\varphi = 0.14$ for two global strain rates. These were $a \approx 103 \text{ s}^{-1}$ and $a \approx 310 \text{ s}^{-1}$ (close to extinction at $a \approx 330 \text{ s}^{-1}$).

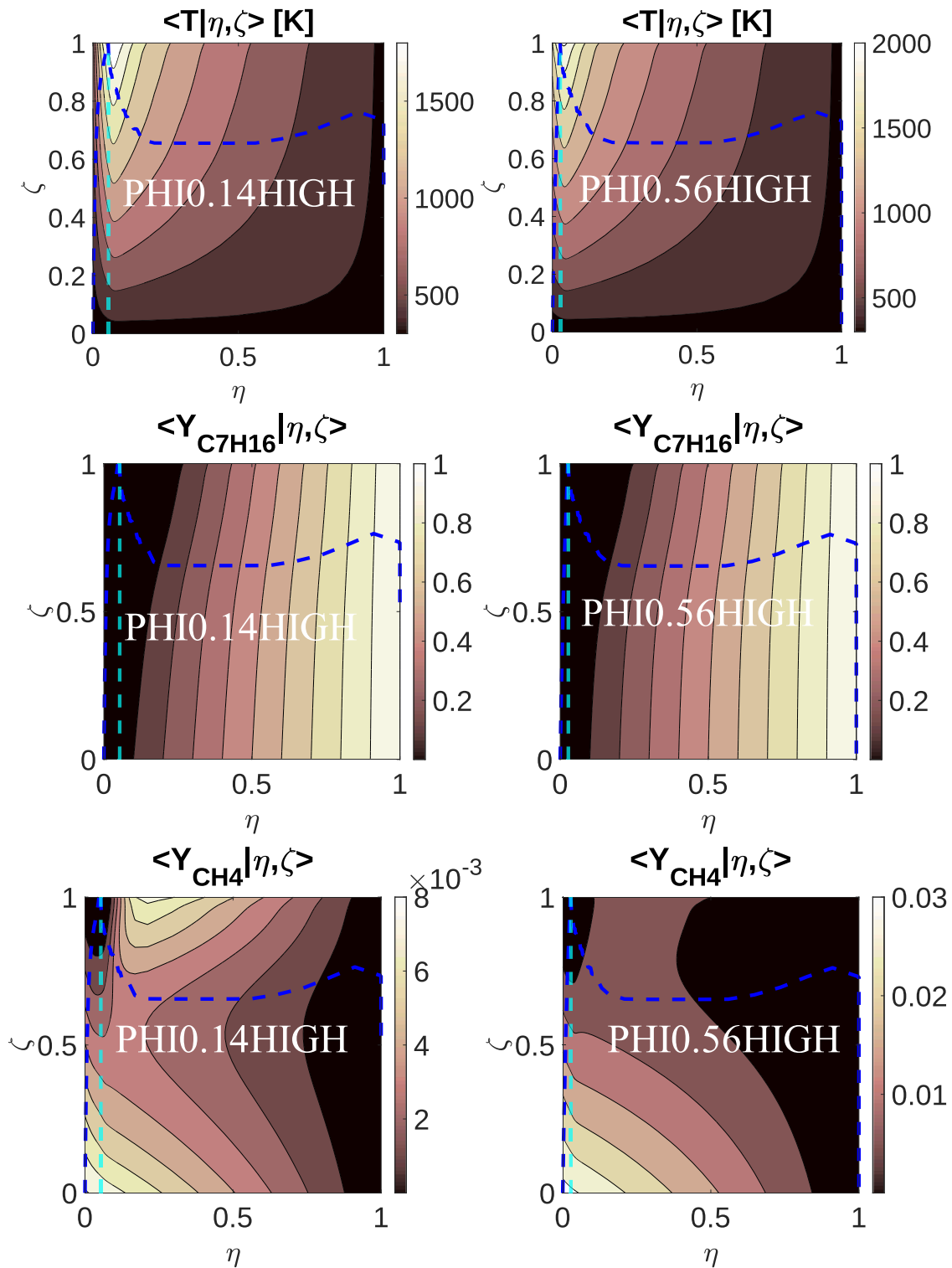


Fig. 7. T , C_7H_{16} and CH_4 and mass fraction distributions in $\eta - \zeta$ space as predicted by the 0D-DCMC solver for flames PHI0.14HIGH and PHI0.56HIGH. The cyan dashed line denotes the stoichiometric composition while the blue dashed line is the $c = fn(\xi)$ curve of the 1D counterflow non-premixed flame predicted by Cantera.

As a increases, the C_7H_{16} oxidation is significantly enhanced, indicated by the more pronounced maximum on the C_7H_{16} side of the HRR profile (Fig. 3a), while the flame becomes thinner. The temperature (Fig. 3b) decreases as expected and the maximum moves towards the C_7H_{16} side. Straining reduces the mass fractions of CH_4 and C_2H_4 on that side (Fig. S3a) and causes the manifestation of O_2 “leakage” in the reaction zone (Fig. S3a). As a increases, the peak CH_2O and HO_2 mass fraction values increase and the OH mass fraction decreases (Fig. S3b),

consistent with incomplete combustion as residence time decreases and extinction is approached.

The 0D-DCMC model usually assumes that $Le=1$. It is thus useful to assess the flame structures using this assumption. The results are shown in Figs 4 and 5 as well as S4 and S5 in the Supplementary Material.

For both a , a temperature underprediction is observed on the C_7H_{16} side (here the region where $\xi > \xi_{st}$) (Fig. 4). The value of that maximum is overpredicted at high a (Fig. 4b) as is the case for the absolute values

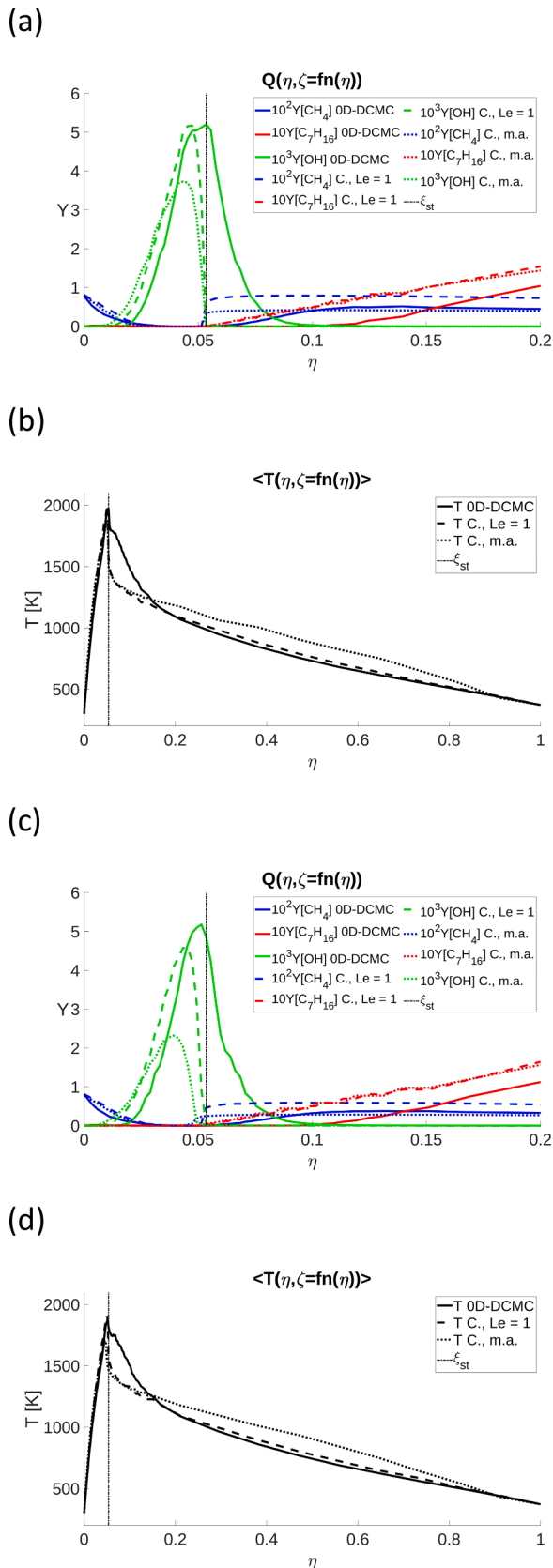


Fig. 8. CH_4 , C_7H_{16} , OH and T distributions along the $c = fn(\xi)$ flame curve for the PHI0.14LOW (a, b) and PHI0.14HIGH (c, d) flame as predicted by the 0D-DCMC model and Cantera (C.) assuming $Le = 1$ or a mixture averaged (m.a.) transport model.

of the local extrema of the HRR profile (Fig. 4b).

There is a significant underprediction of CH_2O mass fraction on the C_7H_{16} side and an overprediction on the CH_4 /air side (here the region where $\xi < \xi_{st}$). High strain enhances the differences in the predictions between the two transport models. The C_7H_{16} mass fraction (Fig. S5) distribution is well predicted while the mass fraction of CH_4 distribution from the $Le = 1$ transport model suffers the opposite distortions compared to CH_2O in terms of modulus (Fig. S5).

Fig. 6 shows the N_ξ , N_c and $N_{\xi c}$ profiles for $\varphi = 0.14$ computed using the two transport models. The scalar dissipation rates in the laminar flame are affected by the transport model. The cross-scalar dissipation is everywhere close to zero in the mixture fraction space, except in the region between stoichiometry and about 0.2 (the point where the HRR drops to zero on the rich side) where it assumes negative values of similar magnitude to the maximum N_ξ .

3.2. 0D-DCMC results

In Figs. 7 and S6 – S7 in the Supplementary Material 2D contour plots of scalars across the whole $\eta - \zeta$ space -the natural output of DCMC- and the $c = fn(\xi)$ curve along which the scalar values are realisable in the Cantera simulations are shown.

Fig. 7 shows that higher temperatures are reached in the higher φ case and indicates that the CH_4 mass fraction is significantly lower at the higher ζ values when the CH_4 /air equivalence ratio is above the lean flammability limit, in agreement with the fact that when this is the case, CH_4 burns in an enhanced premixed mode.

Figs. 8 and S9 – S10 in Supplementary Material depict the 0D-DCMC results along the $c = fn(\xi)$ curve in the $\eta - \zeta$ plane realised in the Cantera predictions employing a $Le = 1$ transport model.

On the CH_4 /air side there is a good agreement as far as the temperature and CH_4 and C_7H_{16} mass fraction distributions are concerned. The greatest discrepancy is observed in the OH mass fraction (Figs. 8a, 8c, S8a and S9a). Although the maximum OH mass fraction from 0D-DCMC is consistent with Cantera, the width of the OH region is slightly wider.

The Cantera predictions for C_7H_{16} mass fraction are effectively independent of the transport model and therefore the underprediction of this species can be solely attributed to the scalar dissipation rates' modelling. It is evident from Figs. 8, S8 and S9 that the $Le = 1$ transport model tends to increase CH_4 on the C_7H_{16} side of the flame. Even though the mass fraction of CH_4 eventually increases away from the stoichiometric region, the tendency of the $Le = 1$ to increase this species' mass fraction in the low φ case (Fig. 8) is cancelled by the evident underprediction attributed the different scalar dissipation rates' profile resulting on a slight overprediction for this species in the rest of the mixture fraction space. In contrast, in the high φ case (Figs. S8 and S9) the effect of the $Le = 1$ transport model dominates. As far as the temperature is concerned, away from the stoichiometric region the effect of the $Le = 1$ assumption dominates once more with the 0D-DCMC model's

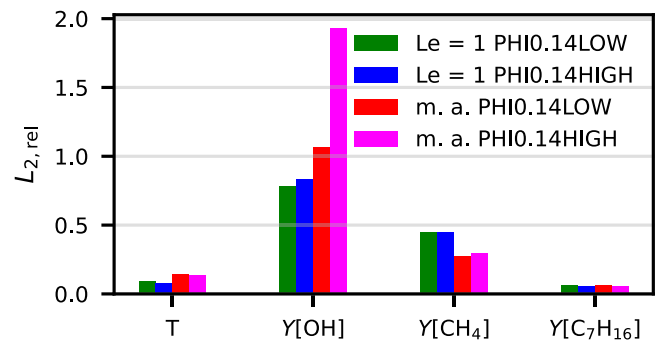


Fig. 9. $L_{2,rel}$ for flames with $\varphi = 0.14$ assuming $Le = 1$ or a mixture averaged (m.a.) transport model.

predictions resembling to a greater extent the Le=1 Cantera results. However, on the C_7H_{16} side of the stoichiometric region, where the temperature predictions of Cantera are again independent of the transport model, for all strain rates considered, there is an overprediction of temperature by the OD-DCMC model.

The relative L_2 norm ($L_{2, rel}$) is defined as:

$$L_{2, rel} = \sqrt{\frac{\sum (\varphi - \varphi_{ref})^2}{\sum \varphi_{ref}^2}} \quad (10)$$

where φ is the predicted value by the model and φ_{ref} is the corresponding reference value. Fig. 9 presents its values for the OD-DCMC predictions of the four quantities under study having as a reference the Cantera predictions for both transport models. It is evident that the model can capture well the temperature and C_7H_{16} mass fraction distributions for all strain rates and moderately that of CH_4 mass fraction. The relative error is greater in the OH mass fraction distribution, particularly at the greater strain rate, but in absolute terms it remains small.

4. Conclusions

Numerical simulations of dual-fuel C_7H_{16} - CH_4 /air laminar counterflow non-premixed flames with a CH_4 /air mixture at an equivalence ratio φ below and above the lean flammability limit have been performed. As φ increased the extinction strain rate manifested a maximum. The flame structures reflected the dual-fuel nature of the flame with two reaction zones, which merged and changed magnitude as the strain rate increased. Assuming Le=1 had a non-negligible effect on the temperature, HRR, intermediate species, and CH_4 mass fraction profiles, but not so much on the profile of C_7H_{16} and decreased the dissipation rates of both mixture fraction and progress variable. Increasing the strain rate enhanced these discrepancies. The OD-DCMC model predicted relatively accurately the distribution of species and temperature on the CH_4 /air side of the flame in mixture fraction space, but with some discrepancies on the C_7H_{16} side. The consumption region of CH_4 and the production region of OH were predicted to be extended and the temperature in the C_7H_{16} side of the stoichiometric region was overpredicted. These are attributed to the different scalar dissipation rates' profiles in the imposed shapes in the OD-DCMC vs the corresponding values and shapes in the actual flame. Away from the stoichiometric region the effect of the Le=1 assumption in general dominates. This is not the case for C_7H_{16} whose mass fraction distribution on the C_7H_{16} side depends mainly on the scalar dissipation rates' modelling. The results suggest that the DCMC formulation offers a reasonable accuracy for the structure of canonical dual-fuel flames and can hence be explored for simulations of turbulent flames. However, care is needed in the scalar dissipation rate modelling.

Novelty and significance statement

The novelty of this research is that it examines at a basic level the structures of dual-fuel flames, which are of interest in various applications and in sequential combustion systems. Further, some of the capabilities and limitations of the DCMC model as a preliminary step to multidimensional LES-DCMC simulations of various dual-fuel systems are explored. The effect of modelling assumptions such as the unity Lewis transport model and the various models used for the scalar dissipation rates, which are very important terms in the DCMC equation, are also explored.

CRediT authorship contribution statement

I. Kestoras: Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **E. Mastorakos:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition.

Declaration of competing interest

The authors have no known competing financial interests or personal relationships that could influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.proci.2025.105966.

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